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Thinking and Programming
Semester- odd
Batch- ALL

Tutorial Assignment: 10

Tutorial title: Functions and Exception Handling

CO Mapping

Name	CO1	CO2	CO3
Iterative Control and Loops	☐	-	-

CONCEPT CLARITY:

Higher Order Functions

A function is called **Higher Order Function** if it contains other functions as a parameter or returns a function as an output i.e, the functions that operate with another function are known as Higher order Functions.

we have three main functions:

1. map()
2. filter()
3. reduce()

The map() function:

The map() function is a higher-order function. As previously stated, this function accepts another function and a sequence of 'iterables' as parameters and provides output after applying the function to each iterable in the sequence. It has the following syntax:

SYNTAX: *map(function, iterables)*

The function is used to define an expression which is then applied to the 'iterables'. User-defined functions and lambda functions can both be sent to the map function.

User-defined functions can be sent to the map() method. The user or programmer is the only one who can change the parameters of these functions.

The filter() function:

The filter() function is used to generate an output list of values that return true when the function is called. It has the following syntax:

SYNTAX: *filter (function, iterables)*

This function like map(), can take user-defined functions and lambda functions as parameters.

Reduce() Function:

reduce() combines elements of the input iterable using a specified function, progressively reducing it to a single value by applying the function to pairs of elements.

Python Modules

As our program grows bigger, it may contain many lines of code. Instead of putting everything in a single file, we can use modules to separate codes in separate files as per their functionality. This makes our code organized and easier to maintain. A file containing Python definitions and statements is called a python module. So naturally, the file name is the module name which is appended with the suffix .py. We use the import keyword to do this.

- 1) Go through the below given python code and write the output.

```
words = ["bay", "cat", "boy", "fan"]  
b_words = list(filter(lambda word: word.startswith("b"), words))  
print(b_words)
```

- 2) The following code will print.....

```
from functools import reduce  
words = ["mary", "had", "a", "little", "lamb"]  
biggest_word = reduce(lambda x, y: x if len(x) > len(y) else y,  
words)  
print(biggest_word)
```

- 3) What does the following code print?

```
l=[1, -2, -3, 4, 5]  
def f1(x):  
    return x<2  
m1=filter(f1, l)  
print(list(m1))
```

- 4) What does the following code print?

```
a = ["Python", "Exceptions", "try and except"]
try:
    for i in range( 4 ):
        print( "Index and element from the array is", i, a[i] )
except:
    print ("Index out of range")
```

5) What does the following code print?

```
import math
def is_positive(num):
    return num >= 0
def sanitized_sqrt(numbers):
    cleaned_iter = map(math.sqrt, filter(is_positive, numbers))
    return list(cleaned_iter)

a=sanitized_sqrt([25, 9, 81, -16, 0])
print(a)
```

6) What does the following code print?

```
def print_year():
    year = 1990
    def print_current_year():
        nonlocal year
        year += 32
        print('Current year is', year)
    print_current_year()

print_year()
```

7) What does the following code print?

```
try:
    True/False
except ZeroDivisionError:
    print('The code gives a ZeroDivisionError.')
print(True/False)
```

8) Find out the error in the below given code?

```
1 dictionary = {'euler_constant': 0.577, 'golden_ratio': 1.618}
2 try:
3     dictionary = dictionary['euler_number']
4 except KeyError:
5     print('This code gives us a KeyError.')
6
7 dictionary = dictionary['euler_number']
8 print(dictionary)
```

Print statement will print which element of the given list

```
import functools
lis=[23,56,78,89,12]
print (functools.reduce(lambda a,b : a if a > b else b,lis))
```

9) What does the following code print?

```
import sys
randomList = ['b', 0, 1]
for entry in randomList:
    try:
        print("value is", entry)
        r = 1/int(entry)
        break
    except Exception as e:
        print("Oops!", e.__class__, "occurred")
```

11) The out of the below given python code is.....

```
def dict_1():
    return locals()
def dict_2():
    program = 'Python'
    return locals()
print('If there is no locals(), it returns an empty dictionary', dict_1())
print('If there is locals(), it returns a dictionary', dict_2())
```

12) What will be the output of the following Python code?

```
from math import factorial
print(math.factorial(5))
```

- a) 120
- b) Nothing is printed
- c) Error, method factorial doesn't exist in the math module
- d) Error, the statement should be: print(factorial(5))

13) Select the correct option for the below given Python code?

```
#mod1 module (mod1.py file)
def change(a):
    b=[x*2 for x in a]
    print(b)

#mod2 module (mod2.py file)
def change(a):
    b=[x*x for x in a]
    print(b)

from mod1 import change
from mod2 import change
#main
s=[1,2,3]
change(s)
```

- a) [2,4,6]
- b) [1,4,9]
- c) [2,4,6] [1,4,9]
- d) Error

14) Write the below given Python code output after execution?

```
a) x = [34, 56]
    print(len(map(str, x)))

b) x = 'abcd'
    print(list(map(list, x)))

c) x = abcd
    print(list(map(list, x)))

d) x = 1234
    print(list(map(list, x)))
```

15) What will be the output of the Python code snippet?

```
def square(n):  
    # This is comment  
    '''Takes in a number n, returns the square of n'''  
    return n**2  
  
print(square.__doc__)
```

16) Find out the output of the following Python code?

```
def fun():  
    a = 45  
    global b  
    b = 6.28  
    print(locals())  
    print(globals())  
  
a = 20  
b = 3.14  
s = 'python programming'  
print(locals())
```

Bonus Question: Write a python program to implement a rock-scissor-paper game using random module.